Chapter 3

AN ECONOMETRIC STUDY OF THE DEMAND FOR ACCESS TO THE INTERNET

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1. INTRODUCTION

The increased popularity and explosive growth of the Internet have been widely tracked. A recent Business Week survey estimates that 40 million people are internet users, which is in line with an April 1997 FIND/SVP survey estimate that there are among 40-45 million adult users. [1] These numbers are up sharply from estimates obtained from surveys conducted in the fall of 1996. A November Louis Harris & Associates' survey, for example, estimated that there were 35 million adult users, while IDC Research estimated that adult users in October 1996 were 31.4 million.[2] PNR and Associates' most recent estimate of households that use the Internet is 16%, which compares with an estimate of 14.8% in 1996. [3] These percentages are consistent with the Harris estimates. Thus, there is ample evidence to support the increased popularity of the Internet and the resulting demand in Internet applications.

1 The starting place for tracking surveys as they relate to the Internet is http://www.nua.ie. The site maintains an extensive Internet survey database. The link is http://www.cyber-atlas.com/demographics.html. The link for FIND/SVP is http://etrg.findsvg.com/internet/interest.html.

2 The links are Op. cit. cyberatlas.com and http://www.idcresearch.com, respectively.

3 The percentages are obtained from the PNR and Market Facts survey, Trends and Linkages in Communications, January 1997. The latter figure is derived from the ReQuest III survey of PNR and Associates.

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Unfortunately, the demand for Internet access is difficult to measure. First, there is the question of what constitutes access. A global definition would include subscriptions to on-line service providers such as AOL and CompuServe, access to the world wide web via any national or local Internet service providers (ISP), access through Web-TV, access via schools and libraries, and access through accounts at work. Second, there is the question of how to obtain accurate measurements of access. And finally, there is the matter of an appropriate framework for modeling Internet demand.

2. MODELING THE DEMAND FOR INTERNET ACCESS

An obvious analogy to pursue in approaching the demand for Internet access is the demand for basic telephone service. While there are a number of similarities between telephone and Internet access, there are also important differences. For both, there is (at least in principle) a distinction between access and usage. For usage to be possible, access is required. Yet the demand for access depends (primarily) upon the benefits derived from usage. In modeling the demand for telephone access, this distinction is taken into account through a two-stage procedure in which usage is modeled (as a function of income, price, and other relevant variables) conditional upon access. Access is then modeled in a probit or logit framework in which the consumer surplus from usage is juxtaposed against the cost of access.[4] In addition, there is a class-of-service choice and brand choice problem embedded in the access decision. That is, the decision is not really do I subscribe to Internet access, but rather which class-of service (flat or measured) and which brand (AOL or MSN) should I purchase if I purchase at all.

Networks are involved in both cases, which makes for modeling complications because of the possible presence of various externalities. First, there is the so-called subscriber or network externality, which makes belonging to a large network more valuable than belonging to a small network. The existence of this externality can cause a network to grow endogenously, even though nothing may be happening to the 'objective' drivers of the system, such as income and prices.[5] While this externality may be a factor in lesser developed countries in which telephone systems are primitive and penetration rates low, it is now probably of only limited

4 For a discussion of these models and their applications, see Taylor (1994).
5 See, for example, Rohlfs (1974) and Littlechild (1975).